

Clinical and Hormonal Predictors of BMI in Women with PCOS: Statistical First Run

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Abstract

We analyze the Kaggle PCOS dataset by Prasoon Kottarathil, with 541 records and 177 PCOS-positive participants. The response variable is BMI. This first run uses descriptive statistics, scatterplots, Pearson correlation, simple linear regression, a Welch two-sample t-test, one-way ANOVA, multiple linear regression, adjusted R^2 , VIF values, residual plots, and sensitivity checks. Fasting insulin and testosterone were not present in the downloaded workbook, so we use available clinical, hormonal, lifestyle, and ultrasound variables.

1 Research Question and Data

Our question is: among women with PCOS, which clinical, hormonal, lifestyle, and ultrasound variables are associated with BMI? The main analysis is restricted to PCOS-positive participants ($n = 177$). We also include a short PCOS vs non-PCOS context table to understand how the analytic subset differs from the rest of the dataset.

2 Community and Research Motivation

The Kaggle dataset page reports substantial public use, including code notebooks and discussions: <https://www.kaggle.com/datasets/prasoonkottarathil/polycystic-ovary-syndrome-pcos>. Recent work using this same dataset groups variables into clinical, laboratory, and ultrasound blocks and studies BMI/cycle subgroups [1]. Explainable-machine-learning work using this dataset highlights follicle counts, AMH, weight gain, skin darkening, hair growth, and FSH/LH-related variables as important for PCOS classification [2]. A recent preprint also notes that the Kaggle diagnosis label lacks verifiable diagnostic criteria, so our results should be framed as exploratory rather than clinical proof [3].

3 Descriptive Statistics

Among PCOS-positive participants, mean BMI was 25.47 with standard deviation 4.40; the median was 25.10. BMI

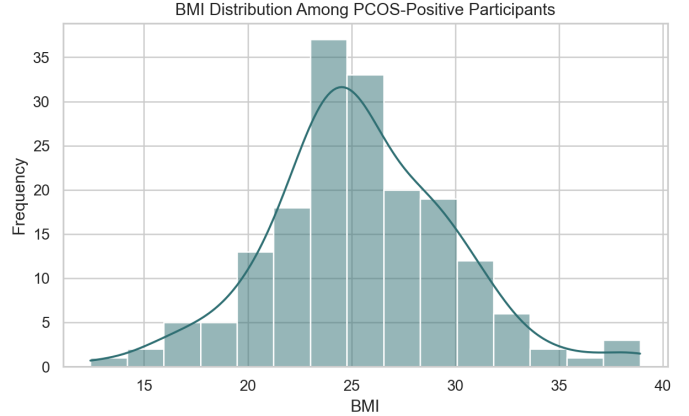


Figure 1: BMI distribution among PCOS-positive participants.

skewness was 0.232, so BMI was not extremely skewed in this subset. Mean AMH was 7.84, and mean waist-to-hip ratio was 0.893.

4 Simple Linear Regression

The simple linear regression used BMI as the response variable and waist-to-hip ratio as the explanatory variable. Pearson correlation was $r = 0.084$ with $p = 0.2678$. The slope estimate was 7.93 BMI units per 1.0 waist-to-hip-ratio unit, but this association was not statistically significant ($p = 0.2678$), and $R^2 = 0.007$.

5 Two-Sample t-Test and ANOVA

Mean BMI was 26.28 for irregular cycles and 24.53 for regular cycles. A Welch two-sample t-test gave $t = -2.70$ and $p = 0.0076$, suggesting higher BMI among participants with irregular cycles before adjustment.

For the one-way ANOVA, we split total follicle count into low, medium, and high groups. The ANOVA was not statistically significant, $F = 0.10$, $p = 0.9039$, so Tukey's HSD was not needed.

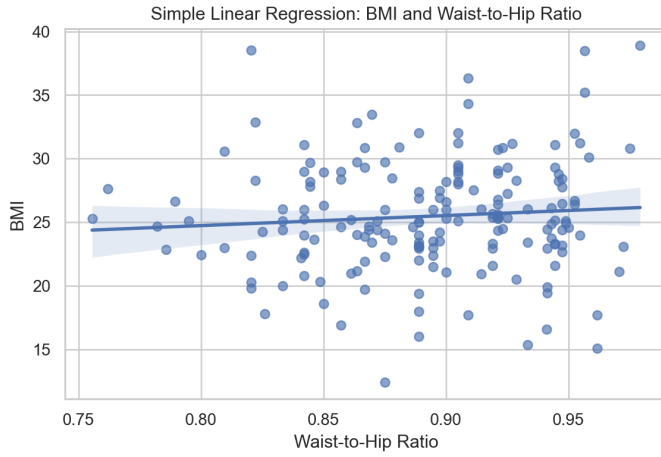


Figure 2: Scatterplot with least-squares regression line for BMI and waist-to-hip ratio.

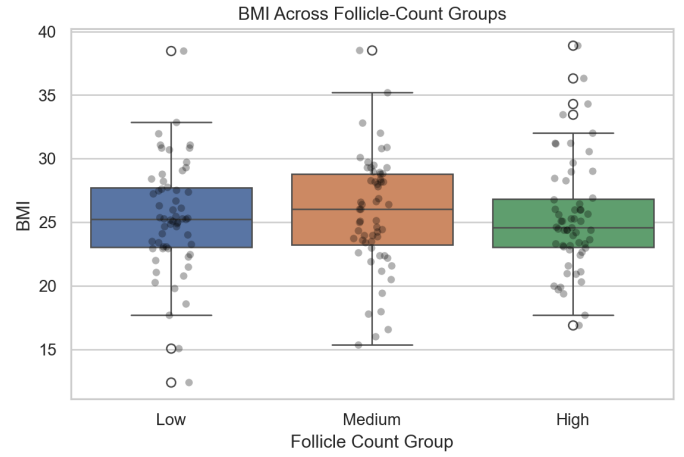


Figure 4: BMI across low, medium, and high follicle-count groups.

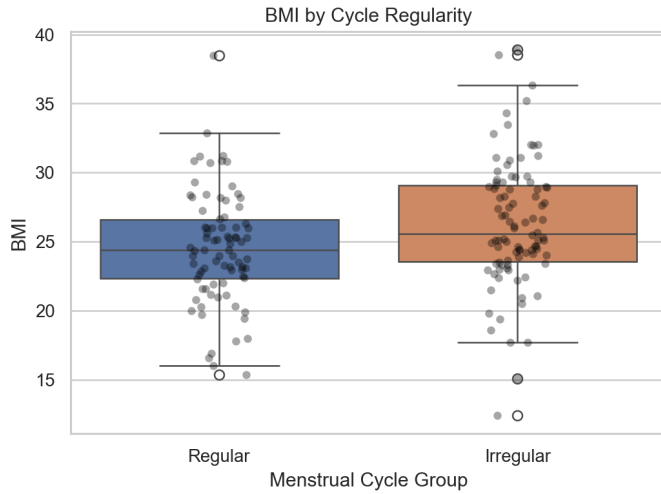


Figure 3: BMI by menstrual cycle regularity.

6 Multiple Regression

The full multiple linear regression included clinical, hormonal, lifestyle, and ultrasound predictors. The full model adjusted R^2 was 0.237; the reduced clinical model adjusted R^2 was 0.260. VIF values for non-intercept predictors were all close to 1, suggesting no major multicollinearity problem.

7 Model Block Comparison

Following the structure used in recent community/research modeling, we compared clinical-only, laboratory-only, ultrasound-only, and combined model blocks using adjusted R^2 . This asks whether easy-to-collect clinical variables explain BMI about as well as more intensive hormone or ultrasound variables.

Table 1: Largest full-model terms by absolute t-statistic.

Term	Estimate	SE	p
weight_gain	4.321	0.686	< 0.001
waist_hip_ratio	11.341	6.969	0.106
fast_food	0.965	0.770	0.212
age	0.072	0.061	0.237
C(cycle_group)[T.Regular]	-0.653	0.617	0.291
tsh	0.102	0.104	0.331
regular_exercise	-0.352	0.698	0.614
lh_fsh_ratio	0.004	0.009	0.671

8 BMI Subgroup Checks

Because recent work stratifies by BMI < 24 vs BMI $\geq$ 24, we used the same threshold as an exploratory subgroup check. Since BMI defines the groups, these results should be interpreted as descriptive comparisons of associated variables rather than causal tests.

9 PCOS vs Non-PCOS Context

Although our main response is BMI among PCOS-positive participants, this table helps the team understand what separates the PCOS-positive and non-PCOS rows in the dataset.

10 Interactions and Sensitivity Checks

We tried a small set of theory-informed interaction terms and sensitivity checks. The goal is not stepwise selection; it is to see whether the main conclusions are stable when we allow simple effect modification or reduce the influence of extreme hormone/BMI values.

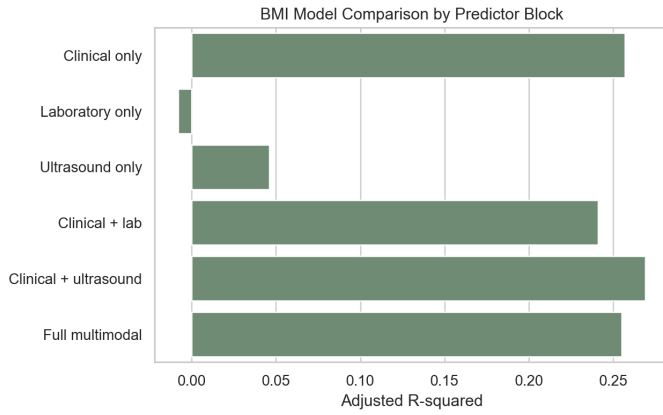


Figure 5: Adjusted R^2 by predictor block.

Table 2: Adjusted R^2 comparison by predictor block.

Model block	n	Adj. R2	AIC
Clinical only	176	0.257	976.200
Laboratory only	176	-0.008	1029.000
Ultrasound only	176	0.046	1019.300
Clinical + lab	176	0.241	984.700
Clinical + ultrasound	176	0.269	978.000
Full multimodal	176	0.255	986.000

11 First Takeaway

The most consistent BMI-related variable in this first run is self-reported weight gain. Cycle irregularity showed a BMI difference in the two-sample t-test, but the coefficient became smaller after adjustment. Waist-to-hip ratio, AMH, LH/FSH ratio, random blood sugar, and ultrasound follicle measures did not show strong independent linear associations with BMI in this PCOS-positive subset.

12 Links to Share

- Kaggle dataset: <https://www.kaggle.com/datasets/prasoonkottarathil/polycystic-ovary-syndrome-pcos>
- AutoPCOS paper using clinical/lab/ultrasound blocks: <https://www.frontiersin.org/journals/endocrinology/articles/10.3389/fendo.2026.1760541/full>
- Explainable ML paper using this Kaggle dataset: <https://www.mdpi.com/2571-5577/6/2/32>
- Staged-modeling preprint discussing label limitations: <https://www.medrxiv.org/content/10.1101/2025.10.27.25338898v2.full-text>
- 2023 international PCOS guideline page: <https://www.monash.edu/medicine/mchri/pcos/guideline>

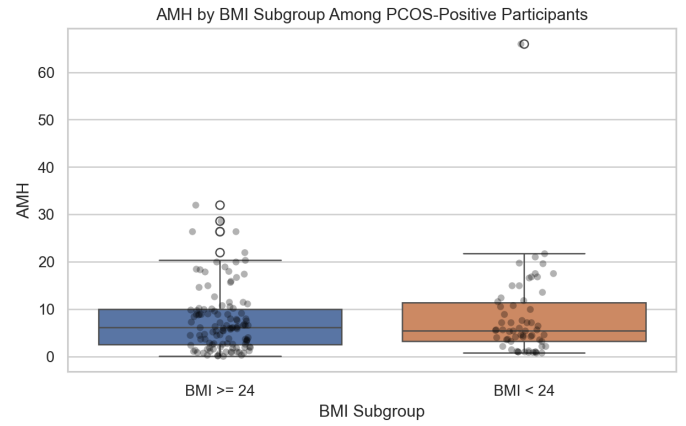


Figure 6: AMH by BMI subgroup among PCOS-positive participants.

Table 3: Exploratory BMI subgroup comparisons among PCOS-positive participants.

Variable	BMI < 24 mean	BMI >= 24 mean	p
AMH	8.265	7.618	0.637
Waist:hip ratio	0.890	0.894	0.523
Total follicles	21.226	20.183	0.399
RBS	99.242	102.158	0.356

References

- [1] Hou et al. AutoPCOS: a stepwise multimodal intelligent framework for polycystic ovary syndrome risk stratification and diagnostic support. *Frontiers in Endocrinology*, 2026.
- [2] Denny et al. A Distinctive Explainable Machine Learning Framework for Detection of Polycystic Ovary Syndrome. *Applied System Innovation*, 2023.
- [3] A Step Toward The Use of AI for Polycystic Ovary Syndrome: Staged Modelling with Uncertainty-Aware Triage and Conformal Prediction with Cost-Efficient Risk. *medRxiv*, 2025.

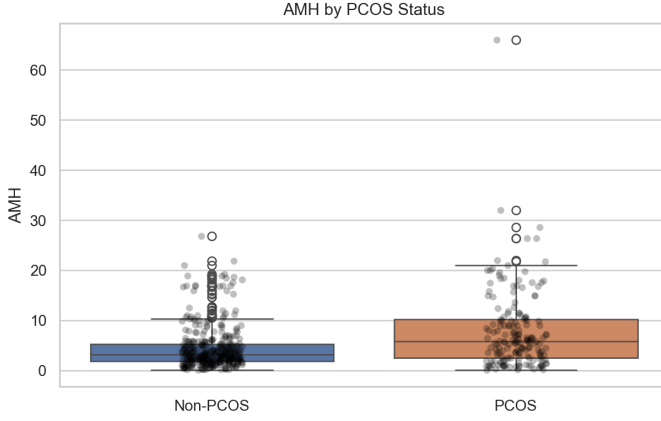


Figure 7: AMH by PCOS status in the full dataset.

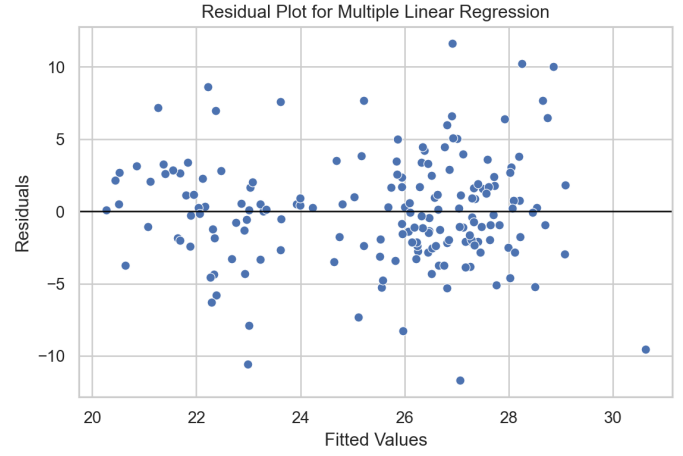


Figure 8: Residual plot for the full multiple linear regression.

Table 4: Exploratory PCOS vs non-PCOS context comparisons.

Variable	PCOS mean	Non-PCOS mean	p
BMI	25.471	23.747	< 0.001
AMH	7.845	4.541	< 0.001
Waist:hip ratio	0.893	0.891	0.774
Total follicles	20.548	8.989	< 0.001
Weight gain	0.684	0.228	< 0.001
Fast food	0.785	0.383	< 0.001
Irregular cycle	0.531	0.154	< 0.001

Table 5: Interaction model comparison.

Model	n	Adj. R2	AIC
Clinical base	176	0.257	976.200
Cycle x weight gain	176	0.261	976.200
Fast food x exercise	176	0.265	975.300
AMH x follicles	176	0.264	985.700

Table 6: Sensitivity checks for model adequacy.

Sensitivity check	n	Adj. R2	Weight gain beta	Weight gain p
Original full model	176	0.255	4.195	< 0.001
Log AMH and log LH/FSH	176	0.258	4.154	< 0.001
Exclude BMI IQR outliers	171	0.240	3.441	< 0.001

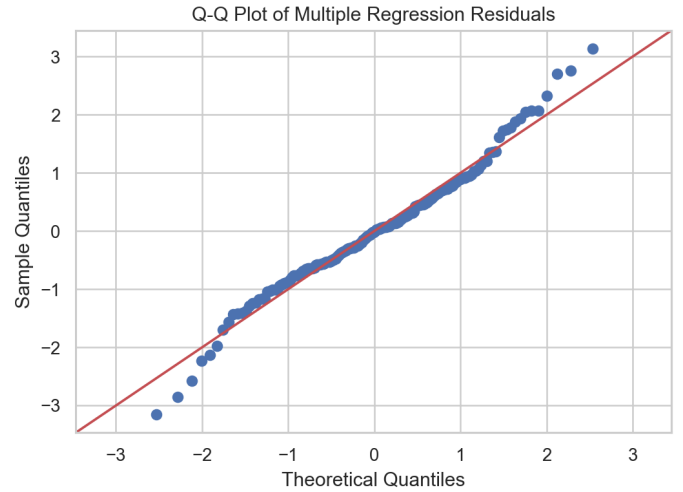


Figure 9: Q-Q plot for the full multiple linear regression residuals.